TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Preparation of Screen

Frame, mesh, photosensitive emulsion, UV exposure and wash-out: building the stencil that flat-screen printing depends on, faults included.

TE-3106 · TEXTILE CHEMISTRY & COLORATION-II SESSIONAL · EXPERIMENT 6 · LEVEL 3 TERM 1

1. Aim and theory

Aim: to prepare a flat printing screen — frame, mesh, coating, exposure and wash-out — capable of printing a sharp single-colour design. A screen is a stencil: mesh threads support a photosensitive emulsion film; UV exposure hardens the emulsion where light passes through the transparent parts of a positive; the unhardened design areas wash out, leaving open mesh that passes paste under the squeegee. Quality is set by mesh count (threads/cm: coarse 20–30 for glitter/thick deposits, standard 40–55 for general work, fine 70+ for halftones), emulsion thickness, and exposure accuracy.

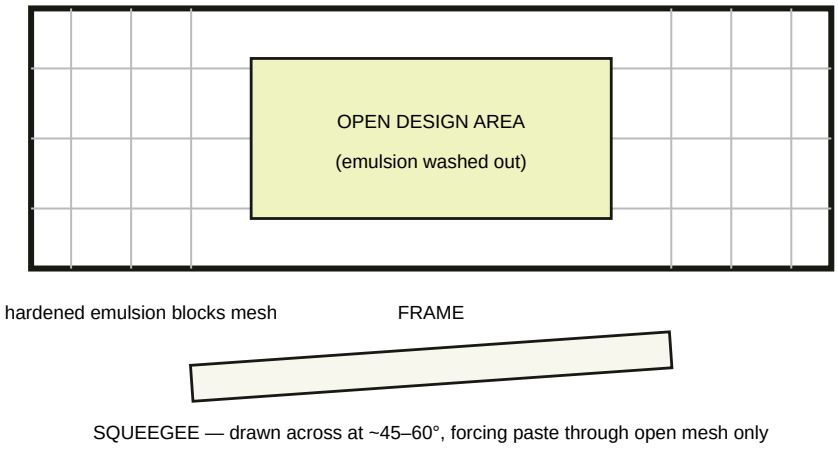


Fig. 1 — Flat screen in plan view with squeegee action: hardened emulsion is the stencil, open mesh is the printing area.

2. Apparatus and materials

- Aluminium or wooden frame (~50 × 70 cm), polyester monofilament mesh 43 threads/cm, staple/pneumatic stretching arrangement or pre-stretched screen.
- Diazo photosensitised emulsion, coating trough, UV exposure unit with vacuum contact, film positive of the design (opaque black on transparent film), wash-out hose with gentle spray, block-out filler, tape.

- Safety: UV unit interlocked/eyewear; emulsion chemicals handled with gloves; darkroom/yellow-light handling before exposure.

3. Procedure

1. **Mesh preparation:** mesh stretched evenly over the frame and fixed (tension uniform — slack mesh prints distorted repeats). Degrease both sides with mesh-prep solution, rinse, dry dust-free: grease causes emulsion pinholes.
2. **Coating:** sensitise emulsion with diazo per maker's ratio; coat print side then squeegee side with the trough in smooth single strokes (typically 2 + 2 coats, wet-on-dry for thickness), in yellow light. Dry horizontally, print side down, in dark air.
3. **Exposure:** place the film positive (reading correctly, emulsion-to-emulsion) on the coated screen; vacuum contact; expose to UV — time by the unit's step-wedge test (typical metal-halide: 2–4 min). Light hardens emulsion everywhere except behind the opaque design.
4. **Wash-out:** wet both sides, wait 1–2 min, spray gently until design areas clear completely and run-off is colourless. Blot, dry, inspect against light: blocked design threads or pinholes are visible at this stage and only this stage cheaply.
5. **Finishing:** block pinholes and edges with filler, tape the frame margins, harden by a short post-exposure. Print a proof on paper, check register marks and edge sharpness, then on fabric.
6. **Reclaiming (end of life):** remove paste, apply emulsion stripper, pressure-wash; mesh returns to step 1.

4. Results and faults (as recorded)

CHECK	RESULT / FAULT AND ITS CAUSE
Design areas fully open	Yes after full wash-out; a first attempt with 40% under-exposure left emulsion scum in fine lines — re-exposed correctly
Pinholes	Three, from dust during coating; blocked with filler
Proof print	Edges sharp; one corner soft where mesh tension was lowest — noted for frame re-stretching

5. Precautions

- Under-exposure washes out detail later on the print table; over-exposure closes fine lines by light undercut — the step-wedge test, not habit, sets time.

- Never let sensitised screens see daylight before exposure; store coated screens dark and use within days.
- Squeegee pressure and angle belong to printing, but a screen with uneven emulsion thickness will print unevenly at any setting — inspect first.

6. Practice (viva preparation)

1. Explain the photochemistry of exposure in one paragraph: what hardens, where, and why the design stays open.
2. Choose mesh counts for: (a) glitter paste, (b) fine halftone, and justify by particle/feature size.
3. A print shows the design plus hundreds of tiny dots elsewhere. Name the fault and its preparation-stage cause.
4. Why is the positive placed emulsion-to-emulsion against the screen?

Answer key

1. UV light activates the diazo sensitiser, crosslinking the emulsion polymer into an insoluble network wherever photons reach it. The film positive's opaque design blocks UV, so emulsion beneath it stays soluble and washes out — those open mesh areas print. Light-struck areas remain as the stencil.
2. (a) Coarse 20–30 threads/cm — large openings pass big glitter particles. (b) Fine 70+ threads/cm — small openings resolve halftone dots and hold thin emulsion detail.
3. Pinholes: dust or grease spots prevented emulsion adhesion/coverage, opening unintended dots; caused at coating/degreasing — cured by clean preparation and block-out filler.
4. Any gap between film image and emulsion lets UV scatter under the opaque edges (undercutting), shrinking fine openings; intimate emulsion-to-emulsion contact keeps edges optically sharp.
